A Trace and Companion-Matrix Solution to the July 2010 IBM *Ponder This* Challenge

Henry Shin

Abstract

We give a structural solution to the July 2010 IBM *Ponder This* challenge asking for a natural number n such that

$$10^9 \mid \operatorname{round}((1+2\cos 20^\circ)^n)\,.$$

The central observation is that

$$\alpha = 1 + 2\cos 20^\circ$$

is a cubic Pisot unit: it is an algebraic integer greater than 1, while its other two conjugates have absolute value less than 1. Consequently, α^n is exponentially close to the integer trace $\operatorname{Tr}(\alpha^n)$. The problem is then reduced to a congruence problem for the trace sequence, or equivalently for the companion matrix of the minimal polynomial of α. A finite-field order computation modulo 2 and 5, followed by a standard lifting argument to 2^9 and 5^9, gives an explicit solution.

Challenge (IBM *Ponder This*, July 2010)**.** The July 2010 IBM *Ponder This* challenge asks for a natural number n such that

$$10^9 \mid \operatorname{round}((1+2\cos 20^\circ)^n)\,.$$

Here $\operatorname{round}(x)$ denotes the nearest integer to x. In the argument below, the number being rounded will always be strictly within $1/2$ of a unique integer, so no convention about half-integers is needed.

1 Introduction and strategy

The problem has a natural elementary recurrence solution. This note gives a more structural version of the same phenomenon. Instead of discovering the recurrence first and then analyzing it, we derive the recurrence as the trace sequence of powers of an algebraic number.

Set

$$\alpha = 1 + 2\cos 20^\circ = 1 + 2\cos\frac{\pi}{9}.$$

The key point is that α is a cubic algebraic integer whose other two conjugates lie strictly inside the unit circle. Thus, if β and γ are the other conjugates, then

$$\alpha^n + \beta^n + \gamma^n$$

is an integer, while $\beta^n + \gamma^n$ rapidly tends to 0. Therefore, for all sufficiently large n,

$$\operatorname{round}(\alpha^n) = \alpha^n + \beta^n + \gamma^n.$$

The rounding in the problem is therefore naturally explained by the trace map.

The remaining task is to force this trace to be 0 modulo 10^9. The trace sequence satisfies a linear recurrence coming from the minimal polynomial

$$X^3 - 3X^2 + 1.$$

Equivalently, it is encoded by the companion matrix

$$A = \begin{pmatrix} 3 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

Since $\det A = -1$, this matrix is invertible modulo every positive integer. The congruence problem becomes a finite group problem: find a power of A that is congruent to the identity modulo 10^9, then use the fact that the trace at index -1 is 0.

We will prove that

$$n = 21{,}699{,}999{,}999$$

works. This value is not claimed to be minimal; the point is to give a clean Galois-theoretic and group-theoretic certificate.

2 The cubic Pisot unit

Let

$$\alpha = 1 + 2\cos\frac{\pi}{9}.$$

We first determine its minimal polynomial and its conjugates.

Lemma 1. *The number α has minimal polynomial*

$$f(X) = X^3 - 3X^2 + 1.$$

Its three conjugates are

$$\alpha = 1 + 2\cos\frac{\pi}{9}, \qquad \beta = 1 + 2\cos\frac{5\pi}{9}, \qquad \gamma = 1 + 2\cos\frac{7\pi}{9}.$$

Moreover,

$$|\beta| < \frac{2}{3}, \qquad |\gamma| < \frac{2}{3}.$$

Proof. We use the identity

$$2\cos(3\theta) = (2\cos\theta)^3 - 3(2\cos\theta).$$

Taking $\theta = \pi/9$, we have $3\theta = \pi/3$, hence

$$(2\cos(\pi/9))^3 - 3(2\cos(\pi/9)) = 2\cos(\pi/3) = 1.$$

Since $\alpha - 1 = 2\cos(\pi/9)$, this gives

$$(\alpha - 1)^3 - 3(\alpha - 1) = 1.$$

Expanding,

$$\alpha^3 - 3\alpha^2 + 1 = 0.$$

Thus α is a root of

$$f(X) = X^3 - 3X^2 + 1.$$

The rational root test shows that f has no rational root, since neither 1 nor -1 is a root. Therefore f is irreducible over $\mathbb{Q}$, and so f is the minimal polynomial of α.

The same trigonometric identity also shows that

$$1 + 2\cos\frac{5\pi}{9} \quad \text{and} \quad 1 + 2\cos\frac{7\pi}{9}$$

are roots of f, because

$$3 \cdot \frac{5\pi}{9} = \frac{5\pi}{3}, \qquad 3 \cdot \frac{7\pi}{9} = \frac{7\pi}{3},$$

and both $\cos(5\pi/3)$ and $\cos(7\pi/3)$ equal $1/2$. These three roots are distinct, so they are precisely the three conjugates of α.

It remains to bound the two smaller conjugates. We compute

$$f\left(-\frac{2}{3}\right) = -\frac{17}{27} < 0, \qquad f(0) = 1 > 0,$$

so one root lies in $(-2/3, 0)$. Also

$$f(0) = 1 > 0, \qquad f\left(\frac{2}{3}\right) = -\frac{1}{27} < 0,$$

so one root lies in $(0, 2/3)$. Finally,

$$f(2) = -3 < 0, \qquad f(3) = 1 > 0,$$

so the remaining root lies in $(2, 3)$. Since

$$\alpha = 1 + 2\cos\frac{\pi}{9} \in (2, 3),$$

the other two conjugates must lie in $(-2/3, 0)$ and $(0, 2/3)$. Hence

$$|\beta| < \frac{2}{3}, \qquad |\gamma| < \frac{2}{3}.$$

$\square$

From the cyclotomic viewpoint, if

$$\zeta = e^{\pi i/9},$$

then

$$\alpha = 1 + \zeta + \zeta^{-1}.$$

Thus α lies in the real cubic subfield of the 18-th cyclotomic field. The three conjugates above correspond to the three embeddings of this real cubic field. This is the Galois-theoretic source of the trace that appears next.

3 The trace sequence and the rounding step

Let
$$K = \mathbb{Q}(\alpha).$$
For $n \in \mathbb{Z}$, define
$$S_n = \mathrm{Tr}_{K/\mathbb{Q}}(\alpha^n) = \alpha^n + \beta^n + \gamma^n.$$

Lemma 2. *For every $n \in \mathbb{Z}$, one has*
$$S_n \in \mathbb{Z}.$$
Moreover, for every $n \geq 4$,
$$\mathrm{round}(\alpha^n) = S_n.$$

Proof. Since f is monic, α is an algebraic integer. Also,
$$\alpha^3 - 3\alpha^2 + 1 = 0$$
implies
$$\alpha^{-1} = 3\alpha - \alpha^2.$$
Thus α is a unit. Hence α^n is an algebraic integer for every $n \in \mathbb{Z}$, and therefore its trace
$$S_n = \mathrm{Tr}_{K/\mathbb{Q}}(\alpha^n)$$
is an integer.

For $n \geq 4$, we have
$$|\alpha^n - S_n| = |\beta^n + \gamma^n| \leq |\beta|^n + |\gamma|^n < 2\left(\frac{2}{3}\right)^n \leq 2\left(\frac{2}{3}\right)^4 = \frac{32}{81} < \frac{1}{2}.$$
Since S_n is an integer, this proves
$$\mathrm{round}(\alpha^n) = S_n$$
for all $n \geq 4$. $\square$

Thus the original problem has been reduced to the following congruence problem:
$$\text{Find } n \geq 4 \text{ such that } S_n \equiv 0 \pmod{10^9}.$$

4 The companion matrix

The trace sequence satisfies a recurrence coming from the polynomial
$$f(X) = X^3 - 3X^2 + 1.$$
Indeed, every root $r \in \{\alpha, \beta, \gamma\}$ satisfies
$$r^3 = 3r^2 - 1.$$
Multiplying by r^{n-3} and summing over the three conjugates gives
$$S_n = 3S_{n-1} - S_{n-3}.$$

It is convenient to encode this recurrence using the companion matrix

$$A = \begin{pmatrix} 3 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

Then

$$\det(TI - A) = T^3 - 3T^2 + 1 = f(T).$$

Therefore the eigenvalues of A over are α, β, γ, and for every $n \geq 0$,

$$\mathrm{tr}(A^n) = \alpha^n + \beta^n + \gamma^n = S_n.$$

We will also need the trace at index -1. Since

$$\det A = -1,$$

the matrix A is invertible over $\mathbb{Z}$, and a direct computation gives

$$A^{-1} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 3 & 0 \end{pmatrix}.$$

Hence

$$\mathrm{tr}(A^{-1}) = 0.$$

This agrees with

$$S_{-1} = \alpha^{-1} + \beta^{-1} + \gamma^{-1} = 0,$$

which also follows from the elementary symmetric relation

$$\alpha\beta + \alpha\gamma + \beta\gamma = 0.$$

So our goal is to find a large positive L such that

$$A^L \equiv I \pmod{10^9}.$$

Then

$$A^{L-1} \equiv A^{-1} \pmod{10^9},$$

and therefore

$$S_{L-1} = \mathrm{tr}(A^{L-1}) \equiv \mathrm{tr}(A^{-1}) = 0 \pmod{10^9}.$$

5 Orders modulo 2 and 5

We now compute suitable orders of A modulo 2 and 5.

Proposition 1. *Modulo 2, the matrix A has order 7. Modulo 5, the matrix A has order 62. In particular,*

$$A^7 \equiv I \pmod 2, \qquad A^{62} \equiv I \pmod 5.$$

Proof. Let
$$f(T) = T^3 - 3T^2 + 1.$$
For a prime p, let $\overline{f}_p$ denote the reduction of f modulo p, and let $\overline{A}_p$ denote the reduction of A modulo p.

First consider $p = 2$. We have
$$\overline{f}_2(T) = T^3 + T^2 + 1.$$
This polynomial has no root in $\mathbb{F}_2$, since
$$\overline{f}_2(0) = 1, \qquad \overline{f}_2(1) = 1.$$
As a cubic with no root, it is irreducible over $\mathbb{F}_2$.

Since $\overline{f}_2$ is the characteristic polynomial of $\overline{A}_2$, and is irreducible of degree 3, it is also the minimal polynomial of $\overline{A}_2$. Thus
$$\mathbb{F}_2[\overline{A}_2] \cong \mathbb{F}_2[T]/(\overline{f}_2) \cong \mathbb{F}_8.$$
Under this isomorphism, $\overline{A}_2$ corresponds to the class $\lambda = T \bmod \overline{f}_2$. Since $\lambda \neq 0, 1$, and $\mathbb{F}_8^\times$ has order 7, we get
$$\lambda^7 = 1$$
and λ has order 7. Therefore
$$\overline{A}_2^7 = I.$$

Now consider $p = 5$. We have
$$\overline{f}_5(T) = T^3 + 2T^2 + 1.$$
Checking the five elements of $\mathbb{F}_5$,
$$\overline{f}_5(0), \overline{f}_5(1), \overline{f}_5(2), \overline{f}_5(3), \overline{f}_5(4) = 1, 4, 2, 1, 2,$$
so $\overline{f}_5$ has no root in $\mathbb{F}_5$. Hence it is irreducible over $\mathbb{F}_5$.

Again,
$$\mathbb{F}_5[\overline{A}_5] \cong \mathbb{F}_5[T]/(\overline{f}_5) \cong \mathbb{F}_{125}.$$
Let $\lambda = T \bmod \overline{f}_5$. The norm of λ from $\mathbb{F}_{125}$ to $\mathbb{F}_5$ is
$$N_{\mathbb{F}_{125}/\mathbb{F}_5}(\lambda) = \lambda^{1+5+25} = \lambda^{31}.$$
On the other hand, since λ is a root of the monic cubic $\overline{f}_5(T) = T^3 + 2T^2 + 1$, the product of its three conjugates is -1. Therefore
$$\lambda^{31} = -1.$$
It follows that
$$\lambda^{62} = 1.$$
Moreover, $\lambda^{31} \neq 1$, and $\lambda \neq -1$, since -1 is not a root of $\overline{f}_5$. The only divisors of 62 are $1, 2, 31, 62$, so λ has order 62. Therefore
$$\overline{A}_5^{62} = I.$$
$\square$

6 Lifting to 2^9 and 5^9

We use a standard lifting lemma.

Lemma 3. *Let p be a prime, and let B be an integer matrix satisfying*

$$B \equiv I \pmod{p}.$$

Then, for every $s \geq 0$,

$$B^{p^s} \equiv I \pmod{p^{s+1}}.$$

Proof. We argue by induction on s. The case $s = 0$ is the hypothesis.

Assume

$$B^{p^s} = I + p^{s+1}C$$

for some integer matrix C. Then

$$B^{p^{s+1}} = \left(B^{p^s}\right)^p = \left(I + p^{s+1}C\right)^p.$$

Expanding,

$$\left(I + p^{s+1}C\right)^p = I + p^{s+2}C + \sum_{j=2}^{p} \binom{p}{j} p^{j(s+1)} C^j.$$

Every term after I is divisible by p^{s+2}. Hence

$$B^{p^{s+1}} \equiv I \pmod{p^{s+2}}.$$

This completes the induction. □

Applying the lemma to $B = A^7$ with $p = 2$, we get

$$A^{7 \cdot 2^8} \equiv I \pmod{2^9}.$$

Applying it to $B = A^{62}$ with $p = 5$, we get

$$A^{62 \cdot 5^8} \equiv I \pmod{5^9}.$$

Now set

$$L = \operatorname{lcm}(7 \cdot 2^8,\ 62 \cdot 5^8).$$

Since $62 = 2 \cdot 31$, this least common multiple is

$$L = 2^8 \cdot 5^8 \cdot 7 \cdot 31 = 217 \cdot 10^8 = 21{,}700{,}000{,}000.$$

Thus

$$A^L \equiv I \pmod{2^9}, \qquad A^L \equiv I \pmod{5^9}.$$

Because 2^9 and 5^9 are coprime, the Chinese remainder theorem gives

$$A^L \equiv I \pmod{10^9}.$$

We can now finish the proof.

Theorem 1. *The integer*

$$\boxed{n = 21{,}699{,}999{,}999}$$

satisfies

$$10^9 \mid \mathrm{round}((1 + 2\cos 20^\circ)^n)\,.$$

Proof. Let

$$L = 21{,}700{,}000{,}000.$$

We have just shown that

$$A^L \equiv I \pmod{10^9}.$$

Therefore

$$A^{L-1} \equiv A^{-1} \pmod{10^9}.$$

Taking traces,

$$\mathrm{tr}(A^{L-1}) \equiv \mathrm{tr}(A^{-1}) \pmod{10^9}.$$

But

$$\mathrm{tr}(A^{L-1}) = S_{L-1}$$

and

$$\mathrm{tr}(A^{-1}) = 0.$$

Hence

$$S_{L-1} \equiv 0 \pmod{10^9}.$$

Since $L - 1 \geq 4$, the rounding lemma gives

$$\mathrm{round}(\alpha^{L-1}) = S_{L-1}.$$

Thus

$$10^9 \mid \mathrm{round}(\alpha^{L-1}).$$

Substituting

$$\alpha = 1 + 2\cos 20^\circ$$

and

$$L - 1 = 21{,}699{,}999{,}999$$

gives

$$10^9 \mid \mathrm{round}\Big((1 + 2\cos 20^\circ)^{21{,}699{,}999{,}999}\Big)\,.$$

Therefore

$$\boxed{n = 21{,}699{,}999{,}999}$$

is a valid answer. $\square$

7 A coarser but very uniform variant

The exact order computation modulo 2 and 5 gives a reasonably small exponent. If one wants an even more uniform group-theoretic certificate, one can avoid computing the exact orders of A modulo 2 and 5.

Since

$$\det A = -1,$$

the reduction of A modulo any prime p lies in

$$\mathrm{GL}_3(\mathbb{F}_p).$$

Therefore the order of A modulo p divides

$$|\,\mathrm{GL}_3(\mathbb{F}_p)| = (p^3-1)(p^3-p)(p^3-p^2).$$

For $p = 2$,

$$|\,\mathrm{GL}_3(\mathbb{F}_2)| = (8-1)(8-2)(8-4) = 168.$$

For $p = 5$,

$$|\,\mathrm{GL}_3(\mathbb{F}_5)| = (125-1)(125-5)(125-25) = 1{,}488{,}000.$$

Thus

$$A^{168} \equiv I \pmod 2, \qquad A^{1{,}488{,}000} \equiv I \pmod 5.$$

The same lifting lemma gives

$$A^{168\cdot 2^8} \equiv I \pmod{2^9}, \qquad A^{1{,}488{,}000\cdot 5^8} \equiv I \pmod{5^9}.$$

Hence

$$M = 2^8 5^8 \cdot 168 \cdot 1{,}488{,}000 = 10^8 \cdot 168 \cdot 1{,}488{,}000 = 24{,}998{,}400{,}000{,}000{,}000$$

satisfies

$$A^M \equiv I \pmod{10^9}.$$

Exactly as before,

$$A^{M-1} \equiv A^{-1} \pmod{10^9},$$

so

$$S_{M-1} \equiv 0 \pmod{10^9}.$$

Since $M - 1 \geq 4$, we again have

$$\mathrm{round}(\alpha^{M-1}) = S_{M-1}.$$

Therefore the coarser group-order argument gives the valid exponent

$$\boxed{n = 24{,}998{,}399{,}999{,}999{,}999}.$$

This version uses only the general finite group bound

$$\mathrm{ord}(A \bmod p) \mid |\,\mathrm{GL}_3(\mathbb{F}_p)|.$$

The sharper proof above replaces those large group orders by the actual orders 7 and 62 of this particular companion matrix modulo 2 and 5.

8 Summary of the mechanism

The proof rests on three linked ideas.

First, the number

$$\alpha = 1 + 2\cos 20^\circ$$

is not just a trigonometric curiosity. It is a cubic algebraic integer with minimal polynomial

$$X^3 - 3X^2 + 1.$$

Its other two conjugates have absolute value less than 1, so powers of α are very close to the integer traces

$$\mathrm{Tr}_{K/\mathbb{Q}}(\alpha^n).$$

This explains the appearance of the rounding function.

Second, the trace sequence

$$S_n = \mathrm{Tr}_{K/\mathbb{Q}}(\alpha^n)$$

satisfies the same recurrence dictated by the minimal polynomial:

$$S_n = 3S_{n-1} - S_{n-3}.$$

The companion matrix packages this recurrence into the identity

$$S_n = \mathrm{tr}(A^n).$$

Third, because the companion matrix is invertible modulo 10^9, its powers are periodic modulo 10^9. Choosing L so that

$$A^L \equiv I \pmod{10^9}$$

forces

$$A^{L-1} \equiv A^{-1} \pmod{10^9}.$$

The trace of A^{-1} is 0, so

$$S_{L-1} \equiv 0 \pmod{10^9}.$$

Since S_{L-1} is exactly the rounded value of α^{L-1}, the desired divisibility follows.